VNUHCM - University of Science
Faculty of Information Technology

CS486 – Lecture 4

Query Languages

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Content

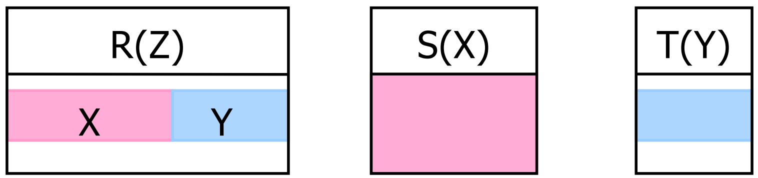
- Introduction
- Define relation schemas
- Manipulate on relation schemas
  - Data modification
  - Data retrieval
    - Basic query
      - On one schemas
      - On two or many schemas
    - **Advanced query**
- Summary

Relational algebra

- Selection
- Projection
- Cartesian product
- Join operation
- Set operations
- **Division operation**
- Other operations
- Modification operations

Division

- Is used to retrieve tuples from *R* that satisfy all tuples from *S*
- Denotation $R \div S$
  - $R(Z)$ and $S(X)$
    - Z is attribute set of R, X is attribute set of S
    - $X \subseteq Z$
- Result is a relation $T(Y)$
  - Has $Y=Z-X$
  - Includes tuples t, if for all $t_S \in S$, there exists a tuple $t_R \in R$ with 2 conditions
    - $t_R(Y) = t$
    - $t_R(X) = t_S(X)$



Division

- Example

R	A	B	C	D	E
	α	a	α	a	1
	α	a	γ	a	1
	α	a	γ	b	1
	β	a	γ	a	1
	β	a	γ	b	3
	γ	a	γ	a	1
	γ	a	γ	b	1
	γ	a	β	b	1

S	D	E
	a	1
	b	1

$R \div S$

A	B	C
α	a	γ
γ	a	γ

Example 25

- Find the SSN of employees who work on **all** the projects

Example 26

- Find the SSN of employees who work for all projects that the department 4 controls

Note

- Express the division operation by the complete set of relational algebra operations

$$Q1 \leftarrow \pi_Y(R)$$

$$Q2 \leftarrow Q1 \times S$$

$$Q3 \leftarrow \pi_Y(Q2 - R)$$

$$T \leftarrow Q1 - Q3$$

Divide operation in SQL

R	A	B	C	D	E
	α	a	α	a	1
	α	a	γ	a	1
	α	a	γ	b	1
	β	a	γ	a	1
	β	a	γ	b	3
	γ	a	γ	a	1
	γ	a	γ	b	1
	γ	a	β	b	1

S	D	E
b_i	a	1
	b	1

R÷S	A	B	C
a_i	α	a	γ
	γ	a	γ

- *R÷S* is a set of values a_i in *R* such that there is no values b_i in *S* that makes the tuple (a_i, b_i) does not exist in *R*

Example 25

- Retrieve the first name of employees who work on ***all*** projects

  - Retrieve the first name of employees such that there is **no** projects that they do **not** work on

  - *R*: WORKS_ON(ESSN, *PNo*)
  - *S*: PROJECT(*PNumber*)
  - $R \div S$: RESULT(ESSN)
  - Join RESULT and EMPLOYEE to retrieve FName

Example 25

- Using NOT EXISTS two times

```
SELECT  R1.A, R1.B, R1.C
FROM    R R1
WHERE   NOT EXISTS (
        SELECT  *
        FROM    S
        WHERE   NOT EXISTS (
                SELECT  *
                FROM    R R2
                WHERE   R2.D=S.D AND R2.E=S.E
                AND R1.A=R2.A AND R1.B=R2.B AND R1.C=R2.C ))
```

Example 25

- Using NOT EXISTS and EXCEPT

```
SELECT  R1.A, R1.B, R1.C
FROM    R R1
WHERE   NOT EXISTS (
        ( SELECT D, E FROM S )
        EXCEPT
        ( SELECT D, E FROM R R2 WHERE R1.A=R2.A
                                  AND R1.B=R2.B
                                  AND R1.C=R2.C ))
```

Example 25 – Relational calculus

- Find employees (SSN) who work on all projects
  - Use the universal quantifier

$$\boxed{\forall t \in R\ (Q(t))}$$

Q is TRUE with all tuples *t* of relation *R*

Example 25

- Find employees (SSN, LNAME, FNAME) who work on all projects

Example 26

- Find employees (SSN, LNAME, FNAME) who work on all projects controlled by the department 4

Example 26

- Find employees (SSN, LNAME, FNAME) who work on all projects controlled by the department 4
  - Use the "implies" operator

$$\boxed{P \Rightarrow Q}$$

If P then Q

Example 26

- Find employees (SSN, LNAME, FNAME) who work on all projects controlled by the department 4

Relational algebra

- Selection
- Projection
- Cartesian product
- Join operation
- Set operations
- Division operation
- **Other operations**
  - Aggregation operators
  - Grouping
  - Outer join
- Modification operations

Aggregation operators

- Input : the collections of values from the DB
- Output : a single value
- Include
  - AVG
  - MIN
  - MAX
  - SUM
  - COUNT

Aggregation operators

- Example

R	A	B
	1	2
	3	4
	1	2
	1	2

SUM(B) = 10

AVG(A) = 1.5

MIN(A) = 1

MAX(B) = 4

COUNT(A) = 4

Grouping

- Is used to consider a relation in groups, corresponding to the value of columns
- Denotation

$$_{G1,\ G2,\ \ldots,\ Gn}\mathcal{F}_{F1(A1),\ F2(A2),\ \ldots,\ Fn(An)}(E)$$

- *E* is relational algebra expression
- *G1, G2, …, Gn* : grouping attributes
- *F1, F2, …, Fn* : aggregation operators
- *A1, A2, …, An* : aggregated attributes

Grouping

- Example

R	A	B	C
	α	2	7
	α	4	7
	β	2	3
	γ	2	10

$\mathcal{F}_{SUM(C)}(R)$

SUM_C
27

${}_A\mathcal{F}_{SUM(C)}(R)$

A	SUM_C
α	14
β	3
γ	10

Example 27

- The number of employees and the average salary of the company

Example 28

- For each department, find the number of employees and the average salary

Example 29

- Find the name of departments that have the largest number of employees

Example 30

- Find the name of employees who work the largest number of projects

Outer join

- Is used to avoid the loss of information
  - A theta join is taken first
  - Then, the tuples that failed to join with any tuple of the other relation are added to the result
- Three cases
  - Left outer join ⟕
  - Right outer join ⟖
  - Left and right outer join ⟗

Example 31

- List out the name of employees and the name of department if they are managers

$$R1 \leftarrow EMPLOYEE ⟕_{SSN=MgrSSN} DEPARTMENT$$

$$RESULT \leftarrow \pi_{FName,\ LName,\ DName}(R1)$$

FName	LName	DName
Franklin	Wong	Research
John	Smith	null
Alicia	Zelaya	null
Jennifer	Wallace	Administration

Example 32

- List out the name of departments and the number of employees of that department

If a department has just been established and the company has not yet arranged employees for it, then what will be the result?

Aggregate functions in SQL

- COUNT
  - *COUNT(*)* : the number of rows
  - *COUNT(<Column_name>)*
    - The number of non-null values of the column
  - *COUNT(DISTINCT <Column_name>)*
    - The number of different and non-null values of the column
- MIN
- MAX
- SUM
- AVG

- These function is in SELECT clause

Example 27

- Find the sum of salary, highest salary, lowest salary and average salary of employees

Example 33

- Find the number of employees in the department "Research"

Example 28

- Find the number of employees for each department

DNo	Num_Emp
5	4
4	3
1	1

SSN	FName	Minit	LName	BDate	Address	Sex	Salary	SuperSSN	DNo
123456789	John	B	Smith	01/09/1965	731 Fondren	Male	30000	333445555	5
333445555	Franklin	T	Wong	12/08/1955	638 Voss	Male	40000	888665555	5
666884444	Ramesh	K	Narayan	09/15/1962	975 Fire Oak	Male	38000	333445555	5
453453453	Joyce	A	English	07/31/1972	5631 Rice	Female	25000	333445555	5
987654321	Jennifer	S	Wallace	06/20/1941	291 Berry	Female	43000	888665555	4
987987987	Ahmad	V	Jabbar	03/29/1969	980 Dallas	Male	25000	987654321	4
999887777	Alicia	J	Zelaya	01/19/1968	3321 Castle	Female	25000	987654321	4
888665555	James	E	Borg	11/10/1937	450 Stone	Male	55000	NULL	1

Grouping

- Syntax

```
SELECT <List_of_columns>
FROM <List_of_ tables>
WHERE <Conditions>
GROUP BY <List_of_grouping_columns>
```

- After grouping
  - Each group will have identical values at grouping attributes

Example 28

- Find the number of employees for each department

Example 34

- For each employee, retrieve the SSN, first name, last name, number of projects as well as total of hours that the employee works on

Example 28

ESSN	PNo	Hours
123456789	1	32.5
123456789	2	7.5
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
888665555	20	20.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
453453453	1	20.0
453453453	2	20.0

Example 34

Example 29

- Find employees who work on two or more projects

ESSN	DNo	Hours
123456789	1	32.5
123456789	2	7.5
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
888665555	20	20.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
453453453	1	20.0
453453453	2	20.0

Eliminated

Conditions on groups

- Syntax

```
SELECT <List_of_columns>
FROM   <List_of_tables>
WHERE <Conditions>
GROUP BY <List_of_grouping_columns>
HAVING <Conditions>
```

Example 35

- Find employees who work on two or more than two projects

Example 36

- Retrieve the department name whose the average salary of employees that is greater than 20000

Discussion

- GROUP BY
  - Attributes in SELECT clause (excepting attributes of aggregate functions) must appear in GROUP BY clause

- HAVING
  - Use aggregate functions in SELECT clause to check a certain condition
  - Just validate the conditions for groups, not a condition for filtering rows
  - After grouping, conditions on groups will be performed

Discussion

- The order of the query execution
  - (0) FROM clause
  - (1) Pick out rows that satisfy conditions in WHERE clause
  - (2) Group these rows into many groups in GROUP BY clause
  - (3) Apply aggregate functions for each group
  - (4) Eliminate groups that do not satisfy conditions in HAVING clause
  - (5) Retrieve values from columns and aggregate functions in SELECT clause
  - (6) ORDER BY clause

Example 37

- Find departments that have the highest average salary

Example 38

- Find three employees who have the highest salary

- If salaries are redundant, then ???

Example 25

- Find the first name of employees who work on all projects

Join conditions in FROM clause

- Equijoin

SELECT <List_of_columns>

FROM R1 [**INNER**] **JOIN** R2 **ON** <Expression>

WHERE <Conditions>

- Outer join

SELECT <List_of_columns>

FROM R1 **LEFT|RIGHT** [**OUTER**] **JOIN** R2 **ON** <Expression>

WHERE <Conditions>

Example 12

- Retrieve the SSN and first name of employees who work for the department "Research"

Example 31

- Retrieve the first and last name of employees, as well as the department name if they are managers

FName	LName	DName
Franklin	Wong	Research
John	Smith	NULL
Alicia	Zelaya	NULL
Jennifer	Wallace	Administration

Example 31

- Retrieve the first and last name of employees, as well as the department name if they are managers

FName	LName	DName
Franklin	Wong	Research
John	Smith	NULL
Alicia	Zelaya	NULL
Jennifer	Wallace	Administration

$$\boxed{\text{EMPLOYEE}} \ \text{join}_{\text{SSN=MgrSSN}} \ \boxed{\text{DEPARTMENT}}$$

Example 31

- Retrieve the first and last name of employees, as well as the department name if they are managers

FName	LName	DName
Franklin	Wong	Research
John	Smith	NULL
Alicia	Zelaya	NULL
Jennifer	Wallace	Administration

DEPARTMENT join EMPLOYEE

MgrSSN=SSN

Example 39

- Retrieve the first and last name of employees, the name of projects that they work on (if any)

$\boxed{\text{WORKS\_ON join PROJECT}}$ join $\boxed{\text{EMPLOYEE}}$

ESSN=SSN

Next week

- Content
  - Advanced query: modification operations
  - Summary
- Read
  - Chapter 5 [The complete book, 2E]
  - Chapter 6, 8 [Fundamentals of Database System, 7E]
- Watch
  - Video clip